

1 Introduction

1.1 Overview

Function

The UDC3200 is a microprocessor-based stand-alone controller. It combines a high degree of functionality and operating simplicity in a 1/4 DIN size controller. This instrument is an ideal controller for regulating temperature and other process variables in numerous heating and cooling applications, as well as in metal working, food, pharmaceuticals, semiconductor, testing and environmental work.

The UDC3200 monitors and controls temperatures and other variables in applications such as environmental chambers, plastic processing machines, furnaces and ovens, and packaging machinery.

Features

- 2 Universal Analog Inputs
- 0.20% Accuracy
- Fast scanning rate (166ms)
- Up to 5 Analog and Digital Output Types
- 2 Digital Inputs
- Math Functions
- Ethernet and Modbus communication
- Infrared PC & Pocket PC configuration
- NEMA4X and IP66 front face protection
- Multilanguage prompts
- 1/4 DIN Size
- Jumper free configuration
- Easily Field Upgradable

Easy to read displays

Bright, dual displays with multi-language prompts (in English, French, German, Spanish, or Italian) make the operator interface easy to read, understand, and operate. Simple keystrokes let you set operating parameters that meet your process control needs.

Analog Inputs

The UDC3200 has two analog inputs with a typical accuracy of $\pm 0.20\%$ of full-scale input and a typical resolution of 16 bits. Both analog inputs are sampled six times per second (every 166 ms).

The first, or Process Variable input, can be one of the various thermocouple, RTD, Radiamatic or linear actuations. Linear actuations have thermocouple, RTD, and Radiamatic transmitter characterization capability as a standard feature. Linear actuations also have square root capability.

The optional second input is isolated and accepts the same actuations as input one, or it provides the Slidewire input for Position Proportional control.

All actuations and characterizations are keyboard configurable. Cold junction compensation is provided for thermocouple type inputs. Upscale, downscale or failsafe sensor break protection is keyboard configurable. A configurable digital filter of 0 to 120 seconds provides input signal damping.

Thermocouple Health - In addition to the standard configurable upscale, downscale or failsafe output burnout selections, the condition of the thermocouple can be monitored to determine if it is good, failing or in danger of imminent failure.

Math Functions

Algorithm—A pre-configured algorithm is available for easy implementation. This includes the capability of using a Ratio and Bias with any input. You can select from the following menu:

Feedforward Summer—Uses either input, followed by a Ratio/Bias calculation, summed directly with the computed PID output value to provide a resultant output to the final control element (standard feature).

Weighted Average—Computes the weighted average of a PV or SP for the control algorithm from two inputs (standard feature).

Feedforward Multiplier—Uses any input, multiplied by the calculated PID output to provide a resultant output which is sent to the final control element (standard feature).

Summer/Subtractor—Will add or subtract inputs with the result used as the derived PV.

Multiplier/Divider—Uses the analog inputs to calculate a derived PV. Available with or without Square Root.

Input High/Low Select—Specifies the PV input as the higher or lower of the two inputs.

Digital Inputs

Two isolated digital inputs are provided for remote dry contact closure to select one of 25 actions. Also, the digital inputs can allow one of six selections to be combined with one of the above selections.

Outputs

Output Types - The UDC3200 may have up to five of the following outputs:

- Current Outputs (4-20 or 0-20 ma)
- Electromechanical Relays (5 amps)
- Solid State Relays (1 amp)
- Dual Electromechanical Relays (2 amps)
- Open Collector Outputs

Alarms

One or two electromechanical alarm relays are available to activate external equipment when preset alarm setpoints are reached. Each of the two alarms can be set to monitor two independent setpoints. Each alarm setpoint can be either high or low alarm. The alarm type can be selected to be either of the inputs, the Process Variable, Deviation, Output, Shed from communications, PV rate of change, or to alarm on manual mode activation or a Current Output Open failure. It can also be used as an On or Off event at the beginning or end of a Ramp/Soak segment. The alarm hysteresis is configurable from 0 to 100% of range.

- Alarms can be configured as latching or non-latching.
- Alarm blocking is also available which allows start-up without alarm energized until after it first reaches the operating region.
- PV rate of change alarm.
- Loop break alarm.
- Timer output reset.
- Diagnostic Alarm

Communications

A communications link is provided between the UDC3200 and a host computer or PLC via the RS422/485 Modbus® RTU or Ethernet TCP/IP communications option. An infrared communication link is also available allowing a non-intrusive configuration of the instrument.

Miscellaneous Features

Auxiliary Output*(optional) - Either or both of the two current outputs can function as an Auxiliary Output which can be scaled from 4-20 ma for 0 to 100% for any range. It can be configured to represent Input 1, Input 2, PV, active Setpoint, Local SP1, Deviation, or the Control Output.

Transmitter Power - This output provides up to 30 volts dc to power a 2-wire transmitter (it requires the use of alarm 2 open collector output selection or the second current output).

Three Local and one Remote Setpoints - Can be configured to provide three Local and one Remote Setpoints, which are selectable either via the keyboard or by Digital Input.

Universal Switching Power - Operates on any line voltage from 90 to 264 Vac 50/60 Hz without jumpers. 24 Vac/dc instrument power is available as an option.

Timer - This standard feature provides a configurable time period of 0 to 99 hours, 59 minutes or units of minutes and seconds. It can be started via the keyboard, alarm 2, or by a digital input. The timer output is Alarm 1, which energizes at the end of the Timer Period. Alarm 1 can be automatically reset. The Timer Period can be changed between each batch. Status is shown on the lower display.

Moisture Protection - The NEMA4X and IP66 rated front face permits use in applications where it may be subjected to moisture, dust, or hose-down conditions.

Setpoint Ramp/Soak Programming (Optional) - Enables you to program and store six Ramp and six Soak segments for setpoint programming. Run or Hold of program is keyboard or remote digital switch selectable.

Setpoint Rate - Lets you define a ramp rate to be applied to any local setpoint change. A separate upscale or downscale rate is configurable. A single setpoint ramp is also available as an alternative.

CE Mark - Conformity with 73/23/EEC, Low Voltage Directive and 89/336/EEC, the EMC Directive as a standard feature.

Approval Body Options - CSA certification is available as an option. UL listing is standard.

Two Sets of Tuning Constants - Two sets of PID parameters can be configured for each loop and automatically or keyboard selected.

Data Security - Five levels of keyboard security protect tuning, configuration, and calibration data, accessed by a configurable 4-digit code. Nonvolatile EEPROM memory assures data integrity during loss of power.

Diagnostic/Failsafe Outputs - Continuous diagnostic routines detect failure modes, trigger a failsafe output value and identify the failure to minimize troubleshooting time.

High Noise Immunity - The controller is designed to provide reliable, error-free performance in industrial environments that often affect highly noise-sensitive digital equipment.

Accutune III™ - This standard feature provides a truly plug and play tuning algorithm, which will, at the touch of a button or through a digital input, accurately identify and tune any process including those with deadtime and integrating processes. This speeds up and simplifies start-up plus allows retuning at any setpoint. The algorithm used is an improved version of the Accutune II™ algorithm found on earlier controllers. Two possibilities are now offered when tuning your process: Fast Tune and Slow Tune.

Fast Tune will tune the process in such a way that the temp is reached faster, a slight overshoot will be allowed.

Slowtune will minimize overshoot, but it will take more time for the process temperature to reach the target setpoint.

Heat/Cool (Duplex Tune) will automatically tune both the heating and cooling sides of the process.

Fuzzy Logic - This standard feature uses fuzzy logic to suppress process variable overshoot due to SP changes or externally induced process disturbances. It operates independently from Accutune III™ tuning. It does not change the PID constants, but temporarily modifies the internal controller response to suppress overshoot. This allows more aggressive tuning to co-exist with smooth PV response. It can be enabled or disabled depending on the application or the control criteria.

1.2 Operator Interface



Figure 1-1 UDC3200 Operator Interface

1.2.1 Function of Displays and Keys

Table 1-1 Function of Displays and Keys

Display Indicators	
3200 Upper display with 4 larger digits shows Process Variable value (normal operation) and special annunciator features. During Configuration, the upper display provides guidance for the operator through prompts (7 – characters) SP 3200 During normal operation, the lower display shows key-selected operating parameters such as Output, Setpoints, Inputs, Deviation, active Tuning Parameter Set, Timer Status, or minutes remaining in a setpoint ramp (4 digits). During configuration, the lower display provides guidance for the operator through prompts (8-characters). ALM Indicates Alarm 1 and/or Alarm 2 conditions exist. DI Indicates Digital Input 1 and/or 2 on.	OUT Indicates Control Relay 1 and/or 2 on. F Or C Indicates either degrees Fahrenheit or Centigrade. MAN Or A Indicates either Manual or Auto mode. SP Indicates Local Setpoint #1. Also, a bar is lighted when the setpoint being used is shown on the lower display.

Keys and Functions

 Function Selects functions within each configuration group.  Setup Scrolls through the configuration groups.  Lower Display Returns Controller to normal display from Set Up mode. Toggles various operating parameters for display.  Increases setpoint or output value. Increases the configuration values or changes functions in Configuration mode groups.  Infrared transceiver	 Man Auto Selects Manual or Auto mode.  SP Select Hold key down to cycle through configured setpoints.  Run Hold Enables Run/Hold of the SP Ramp or Program plus Timer start.  Decreases setpoint or output value. Decreases the configuration values or changes functions in Configuration mode groups.  NEMA4X and IP66 screw attachment (each corner)
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1.3 Process Instrument Explorer Software

Overview

Process Instrument Explorer lets you configure your instrument on a desktop/laptop or Pocket PC. For details see Process Instrument Explorer manual #51-52-25-131.

Features

- Create configurations with intuitive software program running on a Pocket PC, a Desktop or a laptop computer.
- Create/edit configurations live, just connect software to the controller via a communications port.
- Create/edit configurations offline and download to controller later via a communications port.
- Communication types available on every UDC3200:
 - Infrared (standard)
 - RS 485(optional)
 - Ethernet (optional)
- Same port types on UDC2500 and UDC3500 allow interconnectivity.
- This software is available in English, Spanish, Italian, German and French.

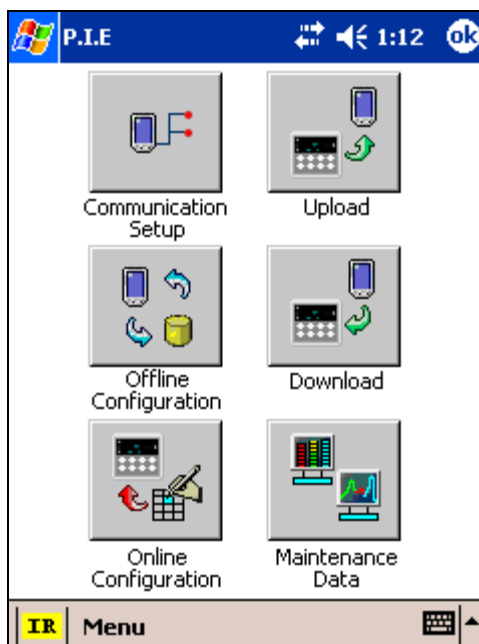


Figure 1-2 Screen capture of Process Instrument Explorer running on a Pocket PC

Infrared communications

The infrared connection provides a non-intrusive wireless connection with the instrument and maintains NEMA4X AND IP66 integrity.

No need to get access to the back of the controller to communicate with the instrument, no need to take your screw driver to wire the communication cable, no wiring mistake possible. You can now duplicate an instrument's configuration, upload or download a new configuration in a matter of seconds, just by pointing your Pocket PC in the direction of the instrument.

It takes just a few seconds to upload a configuration from an instrument. You can then save the configuration file onto your PC or pocket PC for review, modification or archiving. Furthermore, this software also gives you important maintenance information on the controller: instantly, get information on the current operating parameters, digital inputs and alarm status, identify internal or analog input problems.

Question: What if I have several controllers on the same panel? How can I be sure I am communicating with the correct one?

Answer: The infrared port of the controller is normally “off”. You activate the infrared port by pressing any controller's key. You can now communicate. After 4 minutes, the port will be shut down again. Each controller may also be assigned a different communications address.



Figure 1-3 Depiction of infrared communications

1.4 CE Conformity (Europe)

This product is in conformity with the protection requirements of the following European Council Directives: **73/23/EEC**, the Low Voltage Directive, and **89/336/EEC**, the EMC Directive. Conformity of this product with any other “CE Mark” Directive(s) shall not be assumed.

Product Classification: Class I: Permanently connected, panel-mounted Industrial Control Equipment with protective earthing (grounding) (EN61010-1).

Enclosure Rating: This controller must be panel-mounted with the rear terminals enclosed within the panel. The front panel of the controller is rated at NEMA4X and IP66 when properly installed.

Installation Category (Overvoltage Category): Category II (EN61010-1)

Pollution Degree: Pollution Degree 2: Normally non-conductive pollution with occasional conductivity caused by condensation. (Ref. IEC 664-1)

EMC Classification: Group 1, Class A, ISM Equipment (EN61326, emissions), Industrial Equipment (EN61326, immunity)

Method of EMC Assessment: Technical File (TF)

Declaration of Conformity: 51453663

Deviation from the installation conditions specified in this manual, and the special conditions for CE conformity in Subsection 2.1, may invalidate this product's conformity with the Low Voltage and **EMC** Directives.

ATTENTION

The emission limits of EN61326 are designed to provide reasonable protection against harmful interference when this equipment is operated in an industrial environment. Operation of this equipment in a residential area may cause harmful interference. This equipment generates, uses, and can radiate radio frequency energy and may cause interference to radio and television reception when the equipment is used closer than 30 meters (98 feet) to the antenna(e). In special cases, when highly susceptible apparatus is used in close proximity, the user may have to employ additional mitigating measures to further reduce the electromagnetic emissions of this equipment.

WARNING

If this equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.
